Boyer's Casimir Repulsion as Euler Product Truncation: Number-Theoretic Reinterpretation and Three Routes to Experimental Realization

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Boyer (1974) predicted that the Casimir force between a perfectly electrically conducting (PEC) plate and a perfectly magnetically conducting (PMC) plate is repulsive, in contrast to the attractive PEC–PEC case. This prediction remains unverified after fifty years, primarily because no PMC material exists at the relevant (optical–UV) frequencies. We offer a number-theoretic reinterpretation: the PEC–PMC (Dirichlet–Neumann) mode sum $1+3+5+7+\cdots = \zeta_{\neg 2}(-1) = +1/12$ is precisely the Riemann zeta function with the p=2 Euler factor removed, while the PEC–PEC sum $1+2+3+4+\cdots = \zeta(-1) = -1/12$ is the full zeta function. The sign flip is thus the arithmetic operation of *prime muting*: removing the contribution of the smallest prime from the vacuum mode sum. This reinterpretation clarifies why PMC is not required—any physical configuration selecting odd-integer modes suffices. We identify three such configurations that bypass the PMC obstacle: (A) nanoscale open-ended structures providing Neumann boundary conditions without magnetic materials; (B) $\lambda/4$ superconducting resonators in circuit QED, where Lamb shift differences between $\lambda/4$ and $\lambda/2$ resonators probe the vacuum energy sign; and (C) Si-MEMS devices with DN-type geometries, extending the platform of Tang *et al.* (2017). All three are realizable with existing technology. A positive result would constitute the first experimental verification of Boyer's fifty-year-old prediction and the first physical realization of Euler product truncation.

I. INTRODUCTION: A FIFTY-YEAR GAP

In 1974, Boyer showed that the Casimir energy between a perfect electric conductor (PEC, Dirichlet boundary condition for the electric field) and a perfect magnetic conductor (PMC, Neumann boundary condition) is *positive*—opposite in sign to the standard PEC–PEC result [1]. Kenneth *et al.* provided a rigorous proof [2], and Hertzberg *et al.* confirmed it in the piston geometry [3].

Despite this theoretical clarity, **no experiment has ever measured the PEC–PMC Casimir force**. The obstacle is material: PMC requires magnetic permeability $\mu \to \infty$ at the optical and ultraviolet frequencies that dominate the Casimir energy. No natural material satisfies this, and metamaterial approaches fail due to Kramers–Kronig constraints on broadband magnetic response [4].

Several experiments have claimed "Casimir repulsion," but none realizes Boyer's DN sign reversal (Table I).

In every case, the underlying vacuum mode structure is $\sum_{n=1}^{\infty} n^{-s}$: no modes are selectively removed, and the Euler product remains intact. These experiments demonstrate force-direction control or material-contrast effects, *not* vacuum energy sign reversal via mode selection.

The DN Casimir force—where the mode sum is genuinely truncated to $\sum_{n \text{ odd}} n^{-s} = \zeta_{\neg 2}(s)$—has never been experimentally realized.

II. NUMBER-THEORETIC REINTERPRETATION

The Riemann zeta function admits an Euler product over primes:

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}. \tag{1}$$

Removing the p=2 factor yields

$$\zeta_{\neg 2}(s) \equiv (1-2^{-s})\,\zeta(s) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} n^{-s}. \tag{2}$$

The physically relevant case is three-dimensional ($s = -3$), where the Casimir energy density between parallel plates involves $\sum n^3$:

$$\zeta(-3) = 1^3+2^3+3^3+4^3+\cdots = +\tfrac{1}{120}, \tag{3}$$

$$\zeta_{\neg 2}(-3) = 1^3+3^3+5^3+7^3+\cdots = -\tfrac{7}{120}. \tag{4}$$

The sign flip $+1/120 \to -7/120$ is the *defining property* of prime-2 muting in 3D. (In 1D, the analogous flip is $\zeta(-1) = -1/12 \to \zeta_{\neg 2}(-1) = +1/12$.)

The standard 3D Casimir energy density is $\rho_{DD} = -\pi^2\hbar c/(720\,a^4)$, where $720 = 6/\zeta(-3)$. Boyer's result for PEC–PMC is:

- PEC–PEC (DD): all modes $\Rightarrow$ $\rho \propto -\zeta(-3) < 0$ (attractive).
- PEC–PMC (DN): odd modes only $\Rightarrow$ $\rho \propto -\zeta_{\neg 2}(-3) = +7/120 > 0$ (repulsive).

The DN energy density is positive and 7 times the DD magnitude. **Boyer computed $\zeta_{\neg 2}(-3)$ without knowing it.** His electromagnetic boundary-condition calculation and the number-theoretic Euler product truncation are the same mathematical operation, viewed from different angles.

This reinterpretation has a crucial practical consequence: *PMC is not needed.* Any physical configuration whose mode spectrum consists of odd integers realizes $\zeta_{\neg 2}$. The task is not to find a magnetic material, but to find a geometry that selects odd modes.

III. THE $\lambda/4$ REALIZATION

A $\lambda/4$ resonator—one end shorted (Dirichlet), the other open (Neumann)—has mode frequencies

$$f_n = (2n-1)\,\frac{c}{4L}, \quad n = 1, 2, 3, \ldots \tag{5}$$

TABLE I. Prior "Casimir repulsion" experiments and why none constitutes a DN vacuum energy sign reversal.

Experiment	Mechanism	Why not DN repulsion
Munday *et al.* (2009, *Nature*) [5]	Dielectric contrast: gold–bromobenzene–silica. $\varepsilon_1 > \varepsilon_{\text{fluid}} > \varepsilon_2$ (DLP condition).	Mode sum unchanged ($\sum n^{-s}$, all modes present). Repulsion from material contrast, not boundary-condition change. No modes selectively removed.
Tang *et al.* (2017, *Nature Photon.*) [6]	Geometric interplay: T-shaped Si nanostructures slide past each other. Net force vector changes sign at certain lateral displacements.	All pairwise Casimir interactions remain *attractive*. "Repulsion" is the vector sum of multiple attractions in different directions. Mode sum unchanged. No modes removed.
Boyer sphere (1968) [8]	Conducting spherical shell. Casimir self-energy is positive (outward pressure).	Geometry-dependent sign (sphere vs. plates), but mode sum is the full spectrum of the sphere. No prime muting, no DN boundary condition.
Metamaterial proposals [4, 9]	Left-handed or chiral metamaterials engineered to approximate PMC response.	Kramers–Kronig: narrowband $\mu(\omega)$ resonances are smoothed out at imaginary frequencies. Casimir force integral sees no selectivity. Broadband PMC is physically impossible for passive media.

These are the odd integers $1, 3, 5, 7, \ldots$ in units of $c/(4L)$. In 1D, the vacuum energy involves $\zeta(-1)$:

$$E_{\lambda/4} = \frac{\pi\hbar c}{4L}\,\zeta_{\neg 2}(-1) = +\frac{\pi\hbar c}{48L} > 0. \tag{6}$$

Compare the $\lambda/2$ resonator (both ends shorted or both open):

$$E_{\lambda/2} = \frac{\pi\hbar c}{2L}\,\zeta(-1) = -\frac{\pi\hbar c}{24L} < 0. \tag{7}$$

In 3D between parallel plates (area A, separation a), the Casimir energy densities are:

$$\rho_{DD} = -\frac{\pi^2\hbar c}{720\,a^4}, \quad \rho_{DN} = +\frac{7\pi^2\hbar c}{720\,a^4}, \tag{8}$$

with $\rho_{DN}/\rho_{DD} = -7$, directly reflecting $\zeta_{\neg 2}(-3)/\zeta(-3) = -7$. **Opposite signs. No PMC material. No metamaterial. No Kramers–Kronig obstruction.** The open end of a transmission line provides a broadband, dispersion-free Neumann boundary condition for TEM modes.

The obstacle is scale: transmission-line $\lambda/4$ resonators are macroscopic (∼cm), where the Casimir force $\sim 10^{-21}$ N is unmeasurable. We propose three routes around this obstacle.

IV. THREE ROUTES TO EXPERIMENTAL REALIZATION

A. Route A: Nanoscale DN Structures

At the nanoscale ($a \sim 100$ nm), Casimir forces are measurable (∼pN). The challenge is to create DN boundary conditions at this scale.

A Neumann condition for the electromagnetic field corresponds physically to an open aperture, a slot, or a free edge. Candidate structures include:

- Open-ended metallic nanowaveguides (slot arrays in thin metal films).
- Nanoscale coaxial structures with open terminations, fabricable by template-based electrodeposition [7].
- Photonic crystal surfaces engineered to exhibit high-impedance (PMC-like) behavior over a broad bandwidth.

The Casimir force between a PEC plate and such a nano-DN surface can be measured with standard AFM techniques.

B. Route B: Circuit QED ($\lambda/4$ vs. $\lambda/2$)

In circuit QED, a transmon qubit coupled to a superconducting resonator experiences a Lamb shift

$$\delta f = \sum_n \frac{g_n^2}{f_q - f_n}, \tag{9}$$

where f_n are the resonator mode frequencies and g_n the coupling strengths.

A $\lambda/4$ resonator (modes $1, 3, 5, \ldots$) and a $\lambda/2$ resonator (modes $1, 2, 3, \ldots$) of the same fundamental frequency will produce *different* Lamb shifts, because the mode sums in (9) differ. The difference

$$\Delta(\delta f) = \delta f_{\lambda/2} - \delta f_{\lambda/4} = \sum_{n \text{ even}} \frac{g_n^2}{f_q - f_n} \tag{10}$$

isolates the even-mode vacuum fluctuation contribution.

Both $\lambda/4$ and $\lambda/2$ resonators are standard components in superconducting circuit laboratories. The measurement requires a dilution refrigerator ($\sim$10 mK) and a transmon qubit—existing infrastructure at dozens of laboratories worldwide.

C. Route C: MEMS with DN Geometry

Tang *et al.* [6] demonstrated on-chip Casimir force measurement between silicon nanostructures using integrated comb actuators and vibrating-beam force sensors. Their T-shaped geometry produced non-monotonic (but not genuinely DN) forces.

We propose adapting their platform with a DN-type geometry: one surface is a flat PEC (gold-coated silicon), the other is a periodic array of open-ended slots or channels whose mode structure approximates DN boundary conditions. The force is measured as a function of separation a using the same comb-actuator technology.

The key measurement is the scaling exponent α in $F \propto a^{-\alpha}$. Standard PEC–PEC gives $\alpha = 5$ (attractive, well verified). PEC–DN should give $\alpha = 5$ with *opposite sign* (repulsive, never measured).

V. PREDICTIONS

1. **Sign**: The Casimir force in a DN configuration is repulsive. This is Boyer's prediction, now reinterpreted as Euler product truncation $\zeta \to \zeta_{\neg 2}$.
2. **Magnitude**: In 3D, the ratio of DN to DD Casimir energy densities is $\zeta_{\neg 2}(-3)/\zeta(-3) = -7$. The DN repulsive force is 7 times the magnitude of the DD attractive force at the same separation.
3. **Scaling**: Standard QED predicts $F \propto a^{-5}$ for both DD and DN. Any deviation from $\alpha = 5$ in the DN case would indicate physics beyond the standard boundary-effect framework.

VI. DISCUSSION

A. Why This Hasn't Been Done

The PEC–PMC Casimir repulsion has remained unverified for fifty years because the problem was framed as a *materials* problem ("find a PMC"). The number-theoretic reinterpretation reframes it as a *mode selection* problem ("select odd modes"), which can be solved geometrically.

The $\lambda/4$ solution—simply opening one end of a resonator—is arguably the simplest realization of Boyer's prediction. It has not been proposed in this context because (i) $\lambda/4$ resonators are macroscopic where Casimir forces are negligible, and (ii) the connection between DN modes and Euler product truncation was not previously recognized.

B. QED Is Untested in the DN Regime

A critical point: the standard QED prediction $F \propto a^{-5}$ has been confirmed only for DD boundary conditions. **For DN, this prediction has never been experimentally tested.** The belief that QED applies equally to DN rests on theoretical extrapolation, not experimental verification.

DD and DN differ not only in sign but in *topological structure*:

$$K_1(\mathbb{Z}) = \mathbb{Z}/2 \text{ (DD)}, \qquad K_1(\mathbb{Z}[1/2]) = \mathbb{Z}/2 \times \mathbb{Z} \text{ (DN)}. \tag{11}$$

The DN configuration possesses a winding number ($\mathbb{Z}$ factor) absent in DD. QED has been tested in the $K_1 = \mathbb{Z}/2$ sector; the $K_1 = \mathbb{Z}/2 \times \mathbb{Z}$ sector is *terra incognita*.

The precedent for caution is well established: Newtonian gravity is precisely confirmed at low velocities but fails at high velocities (special relativity). A theory's validity in one topological sector does not guarantee its validity in another.

Consequently, this experiment tests not only Boyer's sign prediction but also **whether QED survives its first encounter with a topologically distinct vacuum sector**. If $\alpha = 5$ is confirmed for DN, QED gains validation in a new regime. If $\alpha \neq 5$, the topological structure of the vacuum has physical consequences beyond standard QED.

C. Relation to Broader Program

The Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ decomposes the vacuum into prime-indexed channels. Removing the p=2 channel ($\zeta \to \zeta_{\neg 2}$) is the simplest nontrivial modification. The sign flip $\zeta(-1) \to \zeta_{\neg 2}(-1)$ ($-1/12 \to +1/12$) and $\zeta(-3) \to \zeta_{\neg 2}(-3)$ ($+1/120 \to -7/120$) suggest that prime muting could, in principle, reverse the sign of the cosmological vacuum energy—a necessary condition for

exotic spacetime geometries [10, 11]. Experimental verification of Boyer's sign flip in any of the three proposed routes would be the first step toward testing this broader hypothesis.

D. What This Paper Does Not Claim

We do not claim that prime muting is physically realizable beyond the DN boundary condition. The connection between DN modes and the p=2 Euler factor is a mathematical identity, not a physical mechanism. Whether the arithmetic structure of the Euler product has deeper physical significance is an open question that experiment must address.

VII. CONCLUSION

Boyer's 1974 prediction that PEC–PMC Casimir force is repulsive can be reinterpreted as the statement that removing the p=2 Euler factor from the Riemann zeta function reverses the sign of the regularized vacuum energy. This reinterpretation shifts the experimental challenge from materials (finding a PMC) to geometry (selecting odd modes). We have identified three experimentally viable routes—nanoscale DN structures, circuit QED $\lambda/4$ comparison, and MEMS with DN geometry—each realizable with existing technology. A positive result would simultaneously verify a fifty-year-old theoretical prediction and provide the first physical realization of Euler product truncation in quantum field theory.

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